

1. Implement CUDA program to determine the number of CUDA capable devices in the system. For each device, examine the following device's properties:

- a. Device name
- b. Number of Streaming Multiprocessors
- c. Maximum number of threads per block
- d. Maximum thread dimensions (x,y,z)
- e. Maximum grid dimensions (x,y,z)
- f. The device clock frequency

```
Number of CUDA-capable devices: 1

Device 0
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Device Name: NVIDIA GeForce GT 1030
Number of Streaming Multiprocessors: 3
Maximum Threads per Block: 1024
Maximum Thread Dimensions: (1024, 1024, 64)
Maximum Grid Dimensions: (2147483647, 65535, 65535)
Device Clock Frequency: 1468 MHz
```

2. Given two vectors of length N, implement a program in CUDA that launches two kernel to do the following tasks:

- a) Add two vectors (kernel execution configuration: Block size as N)
- b) Dot product of two vectors (kernel execution configuration: N threads)

```
Vector Addition: C[0] = 9.00
C[1] = 9.00
C[2] = 9.00
C[3] = 9.00
C[4] = 9.00
C[5] = 9.00
C[6] = 9.00
C[7] = 9.00

Dot Product = 120.00
```

3. Implement a CUDA program to compute the Euclidean distance between two randomly initialized arrays A and B, each of length N. Keep the number of threads per block fixed at 256 and dynamically determine the number of blocks required to process all N elements.

```
N = 1000
Threads per block = 256
Number of blocks = 4
Euclidean Distance = 12.630071
```

4. Implement a program in CUDA to process a 1D array containing angles in radians to generate sine of the angle in the output array. Use appropriate function.

```
Angle (radians)    sin(angle)
0.000000           0.000000
0.523599           0.500000
0.785398           0.707107
1.047198           0.866026
1.570796           1.000000
3.141593           -0.000000
4.712389           -1.000000
6.283185           -0.000000
```